Econ 3650: Advanced Econometrics I

Updated course overview for second-year PhD students

This is an updated version of the course. The course keeps the theoretical focus of an advanced PhD econometrics class, but places greater emphasis on hands-on implementation and practical use of the theory. No previous familiarity with nonparametric or semiparametric methods is assumed. Students will learn how identification arguments, estimators, computation, and economic counterfactuals fit together.

Main topics covered

1. Foundations: Identification and Nonparametric Estimation

- Parameters, estimands, estimators, and identification.
- Conditional expectations as infinite-dimensional objects.
- Kernel density estimation and kernel regression.
- Local constant and local linear regression.
- Bias-variance tradeoff, bandwidth choice, and cross-validation.
- Curse of dimensionality and why structure matters.
- Series and sieve estimators: polynomials, splines, and basis functions.

2. Semiparametric Estimation and Modern Methods

- Finite-dimensional parameters with infinite-dimensional nuisance functions.
- Partially linear models and residualization.
- Average derivative estimation and generated regressors.
- Regularization bias and the problem of naive plug-in ML.
- Neyman orthogonality and orthogonal scores.
- Cross-fitting and double/debiased machine learning.
- Applications to treatment effects, policy parameters, and structural parameters.

3. Identification and Partial Identification

- Point identification and the role of assumptions.
- Rank, completeness, moment restrictions, and conditional moments.
- Identified sets, sharp bounds, and outer bounds.
- Missing data, interval data, treatment response, and selection examples.
- Moment inequalities and projections of identified sets.
- Support functions and confidence regions for set-identified parameters.
- Random sets, selections, Aumann expectation, and interval-valued outcomes.

4. Structural Estimation with IO Applications

- What makes a model structural: primitives, equilibrium, and counterfactuals.
- Moment-based estimation, likelihood, simulation, and indirect inference.
- Discrete choice demand: logit, nested logit, substitution patterns, and IIA.
- Differentiated products demand: BLP, random coefficients, and share inversion.
- Price endogeneity, instruments, elasticities, and merger counterfactuals.
- Production function estimation and productivity using proxy-variable methods.
- Dynamic discrete choice and entry games; multiple equilibria and moment inequalities.

Implementation, Course Requirements, and Teaching Goals

Theory, computation, and economic applications in one course

Hands-on and practical implementation

Implementation emphasis in the updated course

- Most modules will include theory-to-code exercises using Julia, Python, or R.
- Students will work with simulated and real datasets to see how estimators behave in practice.
- Problem sets will combine derivations, coding, estimation, and interpretation.
- Computational work will be used to clarify theoretical issues: bias, variance, tuning, identification, and counterfactuals.
- The final project will ask students to apply course tools to an empirical, computational, or theoretical question.

Problem sets

- PS 1: Nonparametric regression and bandwidth choice.
- PS 2: Semiparametric estimation, orthogonalization, and cross-fitting.
- PS 3: Partial identification, bounds, and moment inequalities.
- PS 4: Structural IO estimation and counterfactual analysis.

Grading

- Grading will be based on four problem sets and a final project.
- The problem sets will cover theory, computation, and applications.
- The final project may focus on implementation, replication, identification, or structural estimation.
- The project may use real data, simulated data, or a computational replication of a published method.

Teaching goals

Build a foundation in flexible methods

Nonparametric tools introduce flexible functions, nuisance components, and data-driven estimation.

Learn semiparametric thinking

Students will see how to estimate finite-dimensional targets with flexible first stages and orthogonal methods.

Understand partial identification

The course shows what can still be learned when assumptions are not strong enough for point identification.

Implement advanced estimators

Students will code modern methods rather than treating econometric tools as black boxes.

Connect theory to structural applications

Students will link econometric theory to IO applications such as demand, entry, production, and counterfactual analysis.

The course is designed for students who want to understand both the econometric theory and the practical implementation behind modern empirical work in economics.